NUMBER PLAY

Course Content

Concept explanations • Solved examples • Practice questions with answers

KC	Topic	Skills Covered
KC 1	Number Patterns	Recognise & extend sequences
KC 2	Odd & Even Numbers	Classify integers, apply rules
KC 3	Divisibility Rules	Test for divisors 2, 3, 5, 9, 10
KC 4	Number Sequences	Arithmetic, geometric, oblong series
KC 5	Logical Number Puzzles	Constraint-based reasoning

Knowledge Component 1

Number Patterns

Recognising rules and extending sequences

KC 1 — Number Patterns

What is a Number Pattern?

A number pattern is a sequence of numbers arranged according to a rule. Every pattern has a hidden rule — once you find it, you can predict any term in the sequence.

> **Key Idea:** Look at how each number changes from the previous one. The change itself IS the rule.

Types of Number Patterns

Type	Description	Example
Constant Difference	Same amount added or subtracted each time	2, 5, 8, 11, 14 (rule: +3)
Constant Multiplier	Each term is multiplied by the same number	3, 6, 12, 24, 48 (rule: ×2)
Decreasing	Numbers go down by a fixed amount	50, 45, 40, 35 (rule: −5)
Mixed / Growing	The difference itself changes in a pattern	1, 3, 6, 10, 15 (differences: +2, +3, +4, +5)

How to Find the Rule

1. Write down the differences between consecutive terms.
2. **If the differences are constant** → the rule is to add (or subtract) that number.
3. **If the differences grow or shrink** → look at the differences of the differences (second-order pattern).
4. **If each term is multiplied** → divide consecutive terms to find the multiplier.

Solved Examples

Solved Example: Find the next two terms: 4, 7, 10, 13, ___, ___
Step 1 — Find differences: 7−4=3, 10−7=3, 13−10=3
Step 2 — Identify rule: Constant difference of +3
Step 3 — Extend: 13+3 = 16, 16+3 = 19
Answer: 4, 7, 10, 13, 16, 19

Solved Example: Find the next term: 2, 4, 8, 16, ___
Step 1 — Find multiplier: 4÷2=2, 8÷4=2, 16÷8=2
Step 2 — Rule: Multiply by 2 each time
Step 3 — Extend: 16 × 2 = 32
Answer: 32

Solved Example: Find the next term: 1, 3, 6, 10, 15, ___
Step 1 — Find differences: 3−1=2, 6−3=3, 10−6=4, 15−10=5
Step 2 — Differences grow: The differences are +2, +3, +4, +5, … so next difference is +6
Step 3 — Extend: 15 + 6 = 21
Answer: 21 (These are triangular numbers: 1, 3, 6, 10, 15, 21, …)

Practice Questions

Q1. What is the next term in the sequence: 5, 10, 15, 20, ___?
A) 22
B) 25
C) 30
D) 24

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)

Write down the numbers and look at what is being added each time.	All differences are equal to 5. This is a constant-difference pattern.	Add 5 to the last term. 20 + 5 = 25. The rule is +5 each time.

Remedial Lesson: Constant-Difference Patterns

What is a constant-difference pattern? A sequence where the same number is added (or subtracted) each time.

How to find the rule: Subtract each term from the next: (2nd − 1st), (3rd − 2nd), etc. If all results match, that is your rule.

Example: 3, 7, 11, 15 → differences: 4, 4, 4 → rule is +4.

Apply to this question: 5, 10, 15, 20 → differences: 5, 5, 5 → rule is +5 → next term = 20 + 5 = 25.

✓ **Answer:** B) 25

Explanation: The constant difference is +5. 20 + 5 = 25.

Q2. What is the rule for the sequence: 3, 9, 27, 81?

A) Add 6 each time

B) Multiply by 3 each time

C) Add 18 each time

D) Multiply by 2 each time

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
Try subtracting consecutive terms: 9−3=6, 27−9=18. Are the differences equal?	The differences (6, 18) are NOT equal, so it is not an additive pattern. Try dividing instead: 9÷3=?, 27÷9=?	9÷3=3 and 27÷9=3 and 81÷27=3. Every term is 3 times the previous one.

Remedial Lesson: Multiplicative (Geometric) Patterns

What is a multiplicative pattern? A sequence where each term is multiplied by the same number (called the ratio).

How to identify it: If subtracting gives unequal differences, try dividing: (2nd ÷ 1st), (3rd ÷ 2nd). If equal, that is the ratio.

Example: 2, 6, 18, 54 → 6÷2=3, 18÷6=3, 54÷18=3 → rule is ×3.

Apply to this question: 3, 9, 27, 81 → 9÷3=3, 27÷9=3, 81÷27=3 → rule is multiply by 3 each time.

✓ **Answer:** B) Multiply by 3 each time
Explanation: 9÷3=3, 27÷9=3, 81÷27=3. The constant multiplier is 3.

Q3. Find the missing term: 2, 5, 10, 17, ___, 37

A) 24

B) 25

C) 26

D) 28

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
Write the differences between consecutive terms: 5−2=3, 10−5=5, 17−10=7. Do you notice a pattern in those differences?	The differences are 3, 5, 7 — they increase by 2 each time. What comes after 7? The next difference is 9.	Add the next difference to 17: 17 + 9 = 26. Check: 37 − 26 = 11, and 11 = 9 + 2 ✓

Remedial Lesson: Second-Order (Growing) Differences

What are second-order differences? When the gaps between terms are not constant, look at the gaps between the gaps — those are the second-order differences.

Step 1: Find the first differences: subtract each term from the next.

Step 2: Find the second differences: subtract each first-difference from the next.

Step 3: If second differences are constant, extend them to find the next first-difference, then extend the sequence.

Apply to this question: First diffs: 3, 5, 7, ?, 11. Second diffs: +2, +2, +2. Missing first diff = 9. Missing term = 17 + 9 = 26.

✓ **Answer:** C) 26
Explanation: Differences: +3, +5, +7, +9, +11. The differences increase by +2 each time. 17 + 9 = 26.

Q4. Which sequence shows a decreasing pattern with rule −4?

A) 20, 16, 12, 8, 4

B) 20, 24, 28, 32

C) 20, 15, 10, 5

D) 20, 10, 5, 2.5

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
A decreasing pattern means each term is smaller than the one before it. Which options are decreasing?	Options A, C, and D are decreasing. Now check each: subtract consecutive terms in A. Is the difference always −4?	A: 16−20=−4, 12−16=−4, 8−12=−4, 4−8=−4. All differences are −4.

Remedial Lesson: Decreasing Arithmetic Patterns

What is a decreasing pattern? A sequence where each term is less than the previous term — the constant difference is negative.

How to check: Subtract consecutive terms. If all results equal the same negative number, that is the rule.

Example: 50, 45, 40, 35 → differences: −5, −5, −5 → rule is −5.

Common mistake: Do not confuse rule −5 with rule −10. Always subtract carefully: (next term) − (current term).

Apply to this question: Option A: 16−20=−4 ✓, 12−16=−4 ✓, 8−12=−4 ✓, 4−8=−4 ✓. Rule is −4.

✓ **Answer:** A) 20, 16, 12, 8, 4

Explanation: 16−20=−4, 12−16=−4, 8−12=−4, 4−8=−4. Constant difference of −4.

Q5. What are the next TWO terms in: 100, 90, 81, 73, 66, ___, ___?

A) 60, 55

B) 59, 52

C) 60, 54

D) 60, 53

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
Find the differences: 90−100=−10, 81−90=−9, 73−81=−8, 66−73=−7. What pattern do you see in these differences?	The differences are −10, −9, −8, −7 — they increase by 1 each time (become less negative). What is the next difference after −7?	Next difference = −6, then −5. So: 66 + (−6) = 60, and 60 + (−5) = 55.

Remedial Lesson: Sequences with Changing Differences

Key idea: Not all sequences have constant differences. Sometimes the differences themselves form a pattern.
Step 1: Calculate all first differences (subtracting consecutive terms).
Step 2: Examine those differences — are they arithmetic? Do they increase or decrease by a fixed amount?
Step 3: Extend the difference pattern, then use it to find the next terms.
Apply to this question: Diffs: −10, −9, −8, −7 (each +1). Next diffs: −6, −5. Next terms: 66−6=60, 60−5=55.

✓ **Answer:** A) 60, 55

Explanation: Differences: −10, −9, −8, −7, −6, −5. 66−6=60 and 60−5=55.

Knowledge Component 2

Odd & Even Numbers

Classifying integers and applying arithmetic rules

KC 2 — Odd & Even Numbers

Definitions

Even number: A whole number that is exactly divisible by 2. The last digit is 0, 2, 4, 6, or 8.

Odd number: A whole number that leaves a remainder of 1 when divided by 2. The last digit is 1, 3, 5, 7, or 9.

Quick Classification Rule

You only need to look at the LAST digit of any number to decide whether it is odd or even.

Last Digit	Even or Odd?	Examples
0	Even	10, 100, 2340
2	Even	12, 52, 1002
4	Even	14, 84, 304
6	Even	16, 76, 1006
8	Even	18, 98, 5008
1	Odd	11, 51, 1001
3	Odd	13, 73, 303
5	Odd	15, 85, 1005
7	Odd	17, 47, 107
9	Odd	19, 99, 5009

Arithmetic Rules for Odd & Even

Operation	Result	Example
Even + Even	Even	4 + 6 = 10
Odd + Odd	Even	3 + 5 = 8

Even + Odd	Odd	4 + 3 = 7
Even × Even	Even	4 × 6 = 24
Odd × Odd	Odd	3 × 5 = 15
Even × Odd	Even	4 × 3 = 12

Solved Examples

Solved Example: Classify: 4,738
Look at last digit: 8
8 is in the even set (0,2,4,6,8): 4,738 is EVEN
Verify: 4,738 ÷ 2 = 2,369 — no remainder ✓

Solved Example: Is the sum of 47 and 83 odd or even?
47 is odd, 83 is odd. Odd + Odd = Even
Verify: 47 + 83 = 130 → last digit 0 → Even ✓

Solved Example: A number is multiplied by 6. Can the result be odd?
6 is even. Even × anything = Even
Conclusion: No — the result is always even, regardless of what number is multiplied by 6.

Practice Questions

Q1. Which of the following numbers is ODD?
A) 4,682
B) 3,570
C) 7,913
D) 2,048

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
To identify odd or even, you only need to look at one digit. Which digit is that?	Look only at the LAST digit of each number. Even last digits: 0,2,4,6,8. Odd last digits: 1,3,5,7,9.	A) last digit 2 → even. B) last digit 0 → even. C) last digit 3 → ODD. D) last digit 8 → even.

Remedial Lesson: Identifying Odd and Even Numbers

The rule: A number is even if it ends in 0, 2, 4, 6, or 8. A number is odd if it ends in 1, 3, 5, 7, or 9.

You only need the last digit: No matter how large the number, only the units digit determines odd or even.

Example: 4,782 → last digit 2 → EVEN. 5,937 → last digit 7 → ODD.

Apply to this question: Check last digits: 4,682→2 (even), 3,570→0 (even), 7,913→3 (odd ✓), 2,048→8 (even).

✓ **Answer:** C) 7,913

Explanation: Last digit of 7,913 is 3, which is odd.

Q2. What is the result when an odd number is added to an even number?

A) Always even

B) Always odd

C) Could be either

D) Always zero

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
Try a small example: pick any odd number and any even number and add them. What do you get?	Try 3 + 4 = 7 (odd). Try 7 + 6 = 13 (odd). Does this always hold? Think about why.	An even number ends in 0,2,4,6,8. An odd number ends in 1,3,5,7,9. Adding them gives a last digit that is odd in every case. Odd + Even = always Odd.

Remedial Lesson: Arithmetic Rules for Odd and Even Numbers

Even + Even = Even. Example: 4 + 6 = 10

Odd + Odd = Even. Example: 3 + 5 = 8

Even + Odd = Odd. Example: 4 + 3 = 7 ← this question

Even × Even = Even. Example: 4 × 6 = 24

Odd × Odd = Odd. Example: 3 × 5 = 15

Memory tip: Only Odd + Odd and Odd × Odd 'keep' the odd property in addition and multiplication respectively.

✓ **Answer:** B) Always odd
Explanation: Odd + Even = Odd. Example: 5 + 4 = 9 (odd).

Q3. The product of two numbers is 35. What can you say about the two numbers?

A) Both must be even

B) Both must be odd

C) One is odd, one is even

D) Nothing can be determined

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
35 is an odd number. Can an even number be one of the factors? Think about what even × anything gives.	Even × any number is always even. Since 35 is odd, neither factor can be even.	Both factors must be odd. Check: 5 × 7 = 35. 5 is odd, 7 is odd ✓. Also 1 × 35: both odd ✓.

Remedial Lesson: Multiplication Rules for Odd and Even

Key rule: Even × anything = Even. This means if a product is odd, NEITHER factor can be even.

Odd × Odd = Odd. This is the only multiplication that can produce an odd result.

Odd × Even = Even. Always even.

Even × Even = Even. Always even.

Apply to this question: 35 is odd. For the product to be odd, both numbers must be odd. Confirm: 5 × 7 = 35 ✓ (both odd).

✓ **Answer:** B) Both must be odd
Explanation: 35 is odd. Even × anything = Even. So 35 must be Odd × Odd = Odd.

Q4. How many even numbers are there between 11 and 25 (exclusive)?

A) 5

B) 6

C) 7

D) 8

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
First, what does 'exclusive' mean? It means 11 and 25 are NOT included. List all whole numbers strictly between 11 and 25.	Numbers between 11 and 25 (exclusive): 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24. Now pick only the even ones.	Even numbers in that list: 12, 14, 16, 18, 20, 22, 24. Count them: that is 7 numbers.

Remedial Lesson: Counting Even Numbers in a Range

Understand the range: 'Between 11 and 25 exclusive' means 12, 13, 14, …, 24 (endpoints excluded).

List even numbers: Start at the first even number after 11, which is 12. List every other number: 12, 14, 16, 18, 20, 22, 24.

Count: 12, 14, 16, 18, 20, 22, 24 = 7 even numbers.

Formula shortcut: Count = (last even − first even) ÷ 2 + 1 = (24 − 12) ÷ 2 + 1 = 6 + 1 = 7 ✓

✓ **Answer:** C) 7

Explanation: Even numbers between 11 and 25 (not including 11 or 25): 12, 14, 16, 18, 20, 22, 24 — that is 7.

Q5. If n is odd, which expression is always even?

A) n + 1

B) n + 2

C) n × 1

D) n − 2

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
n is odd. Try n=3 and test each option: A) 3+1=4, B) 3+2=5, C) 3×1=3, D) 3−2=1. Which results are even?	Only A gives an even result for n=3. Now verify for another odd: n=7. A) 7+1=8 (even ✓). So A always gives even.	The rule: Odd + Even = Odd. Adding 1 (even) to an odd number always gives an even number. Answer: n+1.

Remedial Lesson: Algebraic Odd/Even Reasoning

Strategy: Replace n with a specific odd number (e.g. 3 or 5) to test each option. Then confirm with the rule.
Option A — n+1: Odd + 1 = Odd + Even = Even ✓
Option B — n+2: Odd + 2 = Odd + Even = Odd ✗
Option C — n×1: Odd × 1 = Odd × Odd = Odd ✗
Option D — n−2: Odd − 2 = Odd − Even = Odd ✗
Conclusion: Only n+1 is always even when n is odd.

✓ **Answer:** A) n + 1

Explanation: Odd + 1 (even) = Even. n+2 = Odd+Even = Odd. n×1 = Odd. n−2 = Odd−Even = Odd.

Knowledge Component 3

Divisibility Rules

Testing divisibility without performing full division

KC 3 — Divisibility Rules

Why Divisibility Rules?

Divisibility rules allow you to quickly check whether a large number can be exactly divided by a smaller number — without doing long division. This saves time and builds number sense.

The Five Core Rules

Divisor	Rule	Example
2	Last digit is 0, 2, 4, 6, or 8 (even)	348 → last digit 8 → divisible by 2 ✓
3	Sum of all digits is divisible by 3	246 → 2+4+6=12 → 12÷3=4 → divisible ✓
5	Last digit is 0 or 5	785 → last digit 5 → divisible by 5 ✓
9	Sum of all digits is divisible by 9	729 → 7+2+9=18 → 18÷9=2 → divisible ✓
10	Last digit is 0	1,430 → last digit 0 → divisible by 10 ✓

Bonus tip: A number divisible by both 2 AND 3 is also divisible by 6. Check both rules separately.

Solved Examples

Solved Example: Is 4,572 divisible by 3?
Step 1 — Sum the digits: 4 + 5 + 7 + 2 = 18
Step 2 — Is 18 divisible by 3? 18 ÷ 3 = 6 → Yes
Answer: 4,572 is divisible by 3 ✓

Solved Example: Is 8,145 divisible by 9?
Step 1 — Sum the digits: 8 + 1 + 4 + 5 = 18
Step 2 — Is 18 divisible by 9? 18 ÷ 9 = 2 → Yes
Answer: 8,145 is divisible by 9 ✓

Solved Example: Is 7,236 divisible by 6?
Check divisibility by 2: Last digit is 6 (even) → Divisible by 2 ✓
Check divisibility by 3: 7+2+3+6 = 18 → 18÷3 = 6 → Divisible by 3 ✓
Since divisible by both 2 and 3: 7,236 is divisible by 6 ✓

Solved Example: Which of 2, 3, 5, 9, 10 divide 4,320?
By 2: Last digit 0 (even) → Yes ✓
By 3: 4+3+2+0 = 9 → 9÷3 = 3 → Yes ✓
By 5: Last digit 0 → Yes ✓
By 9: Digit sum = 9 → 9÷9 = 1 → Yes ✓
By 10: Last digit 0 → Yes ✓
Answer: 4,320 is divisible by 2, 3, 5, 9, and 10

Practice Questions

Q1. Which number is divisible by 9?
A) 4,561
B) 3,456
C) 7,218
D) 2,345

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
To test divisibility by 9, you do not need to divide the whole number. What shortcut can you use instead?	Add all the digits of each number. A number is divisible by 9 only if its digit sum is divisible by 9.	A) 4+5+6+1=16, not ÷9. B) 3+4+5+6=18, 18÷9=2 ✓. C) 7+2+1+8=18 ✓ — also works! Check both B and C carefully.

Remedial Lesson: Divisibility Rule for 9

The rule: A number is divisible by 9 if the sum of its digits is divisible by 9.

Step 1: Add every digit of the number together.

Step 2: Check if that sum is a multiple of 9 (9, 18, 27, 36, …).

Example: Is 3,456 divisible by 9? 3+4+5+6=18. 18÷9=2. Yes ✓

Note: If the digit sum is large, you can apply the rule again: digit sum of 18 → 1+8=9 ✓

✓ **Answer:** B) 3,456

Explanation: Digit sum of 3,456: 3+4+5+6=18. 18÷9=2. Divisible by 9 ✓

Q2. A number has digit sum 24. Is it divisible by 3?

A) Yes, always

B) No, never

C) Only if the last digit is even

D) Only if the number is even

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
Recall the divisibility rule for 3. What do you need to check about the digits?	The rule for 3: if the digit sum is divisible by 3, the number is divisible by 3. Is 24 divisible by 3?	24 ÷ 3 = 8, no remainder. So YES, any number with digit sum 24 is always divisible by 3.

Remedial Lesson: Divisibility Rule for 3

The rule: A number is divisible by 3 if the sum of all its digits is divisible by 3.

Why the last digit doesn't matter: Unlike divisibility by 2 or 5, divisibility by 3 depends on ALL digits, not just the last one.

Example 1: 246 → 2+4+6=12 → 12÷3=4 ✓ → 246 is divisible by 3.

Example 2: 247 → 2+4+7=13 → 13÷3 leaves remainder 1 → 247 is NOT divisible by 3.

Apply here: Digit sum = 24. 24÷3=8 exactly. So the number is always divisible by 3.

✓ **Answer:** A) Yes, always

Explanation: 24÷3=8. Since the digit sum is divisible by 3, the number itself must be divisible by 3.

Q3. Which of these is divisible by BOTH 5 and 3?

A) 625

B) 450

C) 315

D) Both B and C

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
To be divisible by both 5 AND 3, a number must pass two separate tests. What are those two tests?	Test for 5: last digit must be 0 or 5. Test for 3: digit sum must be divisible by 3. Apply both tests to each option.	A) 625: last digit 5 ✓ (÷5), but 6+2+5=13, 13÷3 has remainder → fails ÷3. B) 450 and C) 315 both pass both tests.

Remedial Lesson: Testing Divisibility by Multiple Numbers

Key idea: Apply each divisibility rule independently. A number must pass ALL rules to be divisible by all of them.

Divisibility by 5: Last digit must be 0 or 5.

Divisibility by 3: Sum of digits must be divisible by 3.

Check 450: Last digit 0 ✓ (÷5). 4+5+0=9, 9÷3=3 ✓ (÷3). Passes both.

Check 315: Last digit 5 ✓ (÷5). 3+1+5=9, 9÷3=3 ✓ (÷3). Passes both.

Check 625: Last digit 5 ✓ (÷5). 6+2+5=13, 13÷3 has remainder ✗. Fails ÷3.

✓ **Answer:** D) Both B and C

Explanation: 450: last digit 0 (÷5 ✓), digit sum 9 (÷3 ✓). 315: last digit 5 (÷5 ✓), digit sum 9 (÷3 ✓).

Q4. Is 7,777 divisible by 3?

A) Yes

B) No

C) Cannot be determined

D) Only if it is even

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
Apply the divisibility rule for 3. What is the sum of the digits of 7,777?	7+7+7+7=28. Now check: is 28 divisible by 3? Divide 28÷3 and check the remainder.	28÷3 = 9 remainder 1. Since the digit sum (28) is not divisible by 3, the number 7,777 is NOT divisible by 3.

Remedial Lesson: Applying the Digit-Sum Rule Carefully

Step 1: Sum the digits: 7+7+7+7 = 28.

Step 2: Check if 28 is divisible by 3: 28 ÷ 3 = 9 remainder 1.

Step 3: Since there is a remainder, 28 is NOT divisible by 3.

Conclusion: 7,777 is NOT divisible by 3.

Tip: Memorise small multiples of 3: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30. If digit sum is not in this list, the number fails the test.

✓ **Answer:** B) No

Explanation: 7+7+7+7=28. 28÷3 leaves remainder 1. 28 is NOT divisible by 3, so 7,777 is not divisible by 3.

Q5. What is the smallest 3-digit number divisible by 9?

A) 108

B) 100

C) 117

D) 99

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
The smallest 3-digit number is 100. Is 100 divisible by 9? Apply the digit-sum rule.	100: 1+0+0=1. 1 is not divisible by 9. The next multiple of 9 after 99 (which is 2-digit) is 9×12=108. Check 108.	108: 1+0+8=9. 9÷9=1 ✓. 108 is divisible by 9 and is a 3-digit number. It is the smallest such number.

Remedial Lesson: Finding the Smallest Number in a Range Divisible by a Divisor

Method 1 — divide and round up:	100 ÷ 9 = 11.1… Round up to 12. 12 × 9 = 108.
Method 2 — digit sum check:	Try 100: 1+0+0=1 ✗. Try 101–107 similarly. Try 108: 1+0+8=9 ✓.
Verify 108 is 3-digit:	108 has three digits ✓. 99 only has two digits, so it is excluded.
Answer:	108 is the smallest 3-digit number divisible by 9.

✓ **Answer:** A) 108

Explanation: Digit sum of 108: 1+0+8=9. 9÷9=1 ✓. 99 is 2-digit. 100: 1+0+0=1, fails. 108 is the smallest 3-digit multiple of 9.

Knowledge Component 4

Number Sequences

Arithmetic, geometric, triangular, square & oblong numbers

KC 4 — Number Sequences

Types of Sequences

Sequence Type	How It Works	General Term	First 5 Terms
Arithmetic	Add a fixed number (common difference d) each time	a + (n−1)d	2, 5, 8, 11, 14 (a=2, d=3)
Geometric	Multiply by a fixed number (common ratio r) each time	a × r^(n−1)	3, 6, 12, 24, 48 (a=3, r=2)
Triangular	Add 1, then 2, then 3, then 4 … to previous term	n(n+1)/2	1, 3, 6, 10, 15
Square	Each term is a perfect square	n^2	1, 4, 9, 16, 25
Oblong	Product of two consecutive integers n×(n+1)	n(n+1)	2, 6, 12, 20, 30

Arithmetic Sequences — Key Formulas

nth term = a + (n−1) × d

where a = first term, d = common difference, n = term position

Sum of n terms = n/2 × (2a + (n−1)d)

Solved Examples

Solved Example: Find the 10th term of: 3, 7, 11, 15, …

Identify a and d: a = 3, d = 7−3 = 4

Apply formula: nth term = 3 + (10−1) × 4 = 3 + 36 = 39
Answer: The 10th term is 39

Solved Example: Find the next term: 2, 6, 12, 20, ___
Find differences: 6−2=4, 12−6=6, 20−12=8
Second differences: +2, +2 → growing by 2
Next difference: 8 + 2 = 10
Next term: 20 + 10 = 30
Alternative method: Terms = 1×2, 2×3, 3×4, 4×5, 5×6 = 30 (oblong numbers)
Answer: 30

Solved Example: In a geometric sequence 5, 15, 45, 135, … what is the 6th term?
Identify a and r: a = 5, r = 15÷5 = 3
Apply formula: 6th term = 5 × 3^(6−1) = 5 × 243 = 1,215
Answer: 1,215

Practice Questions

Q1. What is the 8th term of the arithmetic sequence: 6, 10, 14, 18, …?
A) 34
B) 38
C) 36
D) 40

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
This is an arithmetic sequence. What are the two values you need to use the nth-term formula?	Identify: a (first term) = 6, d (common difference) = 10−6 = 4. The formula is: nth term = a + (n−1) × d. Substitute n=8.	8th term = 6 + (8−1) × 4 = 6 + 7 × 4 = 6 + 28 = 34.

Remedial Lesson: Arithmetic Sequence — nth Term Formula

The formula:	nth term = a + (n − 1) × d
a =	First term of the sequence.
d =	Common difference = (any term) − (previous term).
n =	The position of the term you want to find.
Example:	Sequence 6, 10, 14, 18 → a=6, d=4. 8th term = 6 + 7×4 = 6 + 28 = 34.

✓ **Answer:** A) 34

Explanation: a=6, d=4. 8th term = 6 + (8−1)×4 = 6 + 28 = 34.

Q2. Find the next term in: 2, 6, 12, 20, 30, ___

A) 40

B) 42

C) 44

D) 45

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
Find the differences: 6−2=4, 12−6=6, 20−12=8, 30−20=10. What pattern do the differences follow?	Differences are 4, 6, 8, 10 — they increase by 2. So the next difference is 12. Or try expressing each term as a product.	2=1×2, 6=2×3, 12=3×4, 20=4×5, 30=5×6. The pattern is n×(n+1). Next: 6×7 = 42.

Remedial Lesson: Oblong Numbers n × (n + 1)

What are oblong numbers?	Numbers formed by multiplying two consecutive integers: 1×2, 2×3, 3×4, 4×5, …
The sequence:	2, 6, 12, 20, 30, 42, 56, 72, …
Differences:	+4, +6, +8, +10, +12 — increasing by 2 each time.
Formula:	nth oblong number = n × (n+1).
6th term:	6 × 7 = 42.

✓ **Answer:** B) 42

Explanation: These are oblong numbers n(n+1). Next: 6×7 = 42.

Q3. Which term of the sequence 5, 10, 20, 40, … equals 320?

A) 6th

B) 7th

C) 8th

D) 5th

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
This is a geometric sequence (each term is doubled). Write the formula for the nth term.	nth term = a × r^(n−1) = 5 × 2^(n−1). Set this equal to 320: 5 × 2^(n−1) = 320. Divide both sides by 5.	2^(n−1) = 64. Since 2^6 = 64, we get n−1=6, so n=7. The 7th term is 320.

Remedial Lesson: Geometric Sequence — Finding a Term's Position

The formula: nth term = a × r^(n−1)

a = First term. r = common ratio (divide any term by the previous one).

Set up equation: a × r^(n−1) = target value. Solve for n.

Example: 5 × 2^(n−1) = 320 → 2^(n−1) = 64 → n−1 = 6 → n = 7.

Verify: 7th term: 5, 10, 20, 40, 80, 160, 320 ✓

✓ **Answer:** B) 7th

Explanation: Geometric: a=5, r=2. nth term = 5 × 2^(n−1). 5×2^6=320. n−1=6 → n=7.

Q4. The 5th triangular number is 15. What is the 6th?

A) 18

B) 20

C) 21

D) 22

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
Triangular numbers are formed by adding consecutive integers. What integer was added to get the 5th triangular number (15)?	The 5th triangular number = 1+2+3+4+5 = 15. To get the 6th, you add the next integer: 15 + 6.	15 + 6 = 21. Confirm with formula: n(n+1)/2 = 6×7/2 = 42/2 = 21 ✓

Remedial Lesson: Triangular Numbers
What are triangular numbers? 1, 3, 6, 10, 15, 21, … formed by adding 1, then 2, then 3, etc.
Pattern of differences: +1, +2, +3, +4, +5, +6, …
Formula: nth triangular number = n(n+1) ÷ 2
6th triangular number: 6 × 7 ÷ 2 = 42 ÷ 2 = 21
Alternative: Take the previous term and add the next integer: 15 + 6 = 21.

✓ **Answer:** C) 21

Explanation: Triangular numbers: add the next integer. 15 + 6 = 21. Formula: 6×7/2 = 21 ✓

Q5. Which of these is NOT a perfect square?
A) 144
B) 169
C) 196
D) 200

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
A perfect square is a number whose square root is a whole number. Try taking the square root of each option.	√144=12 ✓, √169=13 ✓, √196=14 ✓. Now try √200. Is it a whole number?	√200 ≈ 14.14 — not a whole number. 200 is between 14^2=196 and 15^2=225, so it is NOT a perfect square.

Remedial Lesson: Perfect Squares
What is a perfect square? A number that equals n^2 for some whole number n. Its square root is a whole number.
Common perfect squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, …
How to test: Take the square root. If it is a whole number, it is a perfect square.
Test 200: √200 ≈ 14.14. Not a whole number. 200 lies between 196 (14^2) and 225 (15^2).
Conclusion: 200 is NOT a perfect square.

✓ **Answer:** D) 200

Explanation: √144=12, √169=13, √196=14. √200 ≈ 14.14 — not a whole number. 200 is not a perfect square.

Knowledge Component 5

Logical Number Puzzles

Constraint-based reasoning and systematic problem-solving

KC 5 — Logical Number Puzzles

What are Logical Number Puzzles?

Logical number puzzles describe a number using a set of clues (constraints). Your task is to find the number that satisfies ALL the clues simultaneously.

> **Strategy:** List all candidates, then eliminate using each clue one at a time. The number left is your answer.

Common Constraint Types

Constraint Type	Example Clue	What to Do
Range	"I am a 2-digit number"	List all numbers 10–99
Divisibility	"I am divisible by 4"	Keep only multiples of 4
Digit properties	"My digits sum to 11"	Check digit sums
Odd / Even	"I am odd"	Keep only odd numbers
Comparison	"I am greater than 50"	Eliminate numbers ≤ 50
Remainder	"I leave remainder 2 when divided by 5"	Last digit must be 2 or 7

Step-by-Step Solving Method

5. Read all clues first — identify the most restrictive one.
6. List all numbers satisfying the most restrictive clue.
7. Apply remaining clues one by one, crossing out numbers that fail.
8. Verify your final answer satisfies ALL clues.

Solved Examples

Solved Example: I am a 2-digit number. My digits sum to 9. I am divisible by 3. I am even. What am I?
2-digit: Numbers 10–99
digit sum = 9: 18, 27, 36, 45, 54, 63, 72, 81, 90
divisible by 3: All of the above already are (digit sum 9 guarantees ÷3)
even: 18, 36, 54, 72, 90
Answer: Multiple answers satisfy all clues: 18, 36, 54, 72, 90. If the puzzle expects one answer, check for an additional clue.

Solved Example: I am between 30 and 50. I am divisible by 7. My digit sum is even. What am I?
Between 30 and 50: 31, 32, … 49
Divisible by 7: 35, 42, 49
Digit sum is even: 35: 3+5=8 ✓ | 42: 4+2=6 ✓ | 49: 4+9=13 ✗
Remaining: 35 and 42
Note: If the puzzle expects a unique answer, one more clue is needed. With these three clues: 35 or 42.

Solved Example: Find a 2-digit number whose digits multiply to 18 and sum to 9.
Digit pairs that multiply to 18: (2,9), (3,6), (6,3), (9,2)
Check digit sum = 9: 2+9=11 ✗ | 3+6=9 ✓ | 6+3=9 ✓ | 9+2=11 ✗
Valid numbers: 36 and 63
Answer: 36 or 63

Practice Questions

Q1. I am a 2-digit number. I am divisible by both 3 and 5. I am less than 50. What could I be?
A) 15
B) 30

C) 45

D) All of A, B and C

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
A number divisible by both 3 and 5 must be divisible by what larger number? List its 2-digit multiples.	Divisible by 3 AND 5 → divisible by 15. 2-digit multiples of 15: 15, 30, 45, 60, 75, 90. Which are less than 50?	15 < 50 ✓, 30 < 50 ✓, 45 < 50 ✓, 60 > 50 ✗. All three (15, 30, 45) satisfy every clue, so answer is D.

Remedial Lesson: Systematic Elimination with Multiple Constraints

Step 1: Identify the most restrictive clue. 'Divisible by both 3 and 5' → divisible by 15.

Step 2: List all 2-digit multiples of 15: 15, 30, 45, 60, 75, 90.

Step 3: Apply 'less than 50': keep 15, 30, 45. Remove 60, 75, 90.

Step 4: All three remaining values are 2-digit ✓, divisible by 15 ✓, less than 50 ✓.

Answer: All of A, B, and C satisfy all constraints.

✓ **Answer:** D) All of A, B and C

Explanation: Divisible by 3 and 5 means divisible by 15. 2-digit multiples of 15 less than 50: 15, 30, 45 — all satisfy every clue.

Q2. A number leaves remainder 1 when divided by 2, and remainder 1 when divided by 3. The number is between 10 and 20. What is the smallest such number?

A) 13

B) 17

C) 11

D) 19

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
Remainder 1 when divided by 2 means the number is odd. List all odd numbers between 10 and 20.	Odd numbers between 10 and 20: 11, 13, 15, 17, 19. Now check which leave remainder 1 when divided by 3.	11÷3=3r2 ✗. 13÷3=4r1 ✓. 15÷3=5r0 ✗. 17÷3=5r2 ✗. 19÷3=6r1 ✓. Smallest is 13.

Remedial Lesson: Remainder Constraints

What does 'remainder 1 when divided by 2' mean? The number is odd (all odd numbers leave remainder 1 when divided by 2).

What does 'remainder 1 when divided by 3' mean? The number is of the form 3k+1: 1, 4, 7, 10, 13, 16, 19, 22, …

Combine constraints: Must be odd AND of the form 3k+1, between 10 and 20.

List and check: Odd numbers in range: 11, 13, 15, 17, 19. Check ÷3 remainder: 13 ✓, 19 ✓.

Smallest: 13

✓ **Answer:** A) 13

Explanation: Remainder 1 ÷2 → odd: 11,13,15,17,19. Remainder 1 ÷3: 13 (13÷3=4r1✓), 19 (19÷3=6r1✓). Smallest is 13.

Q3. The sum of two numbers is 20 and their product is 96. What are the two numbers?

A) 8 and 12

B) 6 and 14

C) 4 and 16

D) 10 and 10

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
You have two clues: sum = 20, product = 96. Try each option and check the SUM first.	All options sum to 20. Now check the product for each: A) 8×12=?, B) 6×14=?, C) 4×16=?, D) 10×10=?	A) 8×12=96 ✓. B) 6×14=84 ✗. C) 4×16=64 ✗. D) 10×10=100 ✗. Answer: A.

Remedial Lesson: Sum and Product Puzzles

Strategy: List all pairs that add up to the given sum, then check which pair also gives the required product.

Pairs that sum to 20: (1,19), (2,18), (3,17), (4,16), (5,15), (6,14), (7,13), (8,12), (9,11), (10,10).

Check products: 8×12=96 ✓. All others give a different product.

Verify: 8+12=20 ✓ and 8×12=96 ✓

Algebraic shortcut: x+y=20 and xy=96. y=20−x. x(20−x)=96 → $x^2-20x+96=0$ → (x−8)(x−12)=0 → x=8 or 12.

✓ **Answer:** A) 8 and 12

Explanation: 8+12=20 ✓ and 8×12=96 ✓. Check others: 6×14=84 ✗, 4×16=64 ✗, 10×10=100 ✗.

Q4. I am a 3-digit palindrome (reads the same forwards and backwards). My digit sum is 15. My first digit is 6. What am I?

A) 636

B) 696

C) 666

D) 663

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
A 3-digit palindrome has the form ABA (first and last digit are the same). If the first digit is 6, what is the last digit?	The form is 6X6. Now use the digit sum: 6 + X + 6 = 15. Solve for X.	6 + X + 6 = 15 → 12 + X = 15 → X = 3. The number is 636.

Remedial Lesson: Palindrome Number Puzzles

What is a palindrome? A number that reads the same forwards and backwards. E.g. 121, 343, 636.

3-digit palindrome form: ABA — first and last digit are identical, middle digit can be anything.

Use the constraints: First digit = 6 → form is 6B6. Digit sum = 15: 6 + B + 6 = 15 → B = 3.

Answer: 636. Check: reads same backwards ✓, digit sum 6+3+6=15 ✓, first digit 6 ✓.

✓ **Answer:** A) 636

Explanation: 3-digit palindrome starting with 6: form is 6X6. Digit sum: 6+X+6=15 → X=3. Number: 636.

Q5. A number is multiplied by 3 and then 7 is added. The result is 34. What is the original number?

A) 7

B) 8

C) 9

D) 11

Hint 1 (after attempt 1)	Hint 2 (after attempt 2)	Hint 3 (after attempt 3)
Work backwards from the result. The last operation was '+7'. What do you do to reverse it?	Reverse '+7' by subtracting 7: 34 − 7 = 27. Now reverse '×3' by dividing: 27 ÷ 3 = ?	27 ÷ 3 = 9. The original number is 9. Verify: 9 × 3 = 27, then 27 + 7 = 34 ✓.

Remedial Lesson: Working Backwards (Inverse Operations)
Key idea: To find the original number, undo each operation in reverse order.
The operations: Original → ×3 → +7 → 34.
Reverse step 1: Undo '+7': 34 − 7 = 27.
Reverse step 2: Undo '×3': 27 ÷ 3 = 9.
Verify: 9 × 3 + 7 = 27 + 7 = 34 ✓
Algebraic version: 3n + 7 = 34 → 3n = 27 → n = 9.

✓ **Answer:** C) 9

Explanation: Let the number be n. 3n + 7 = 34 → 3n = 27 → n = 9. Check: 3×9+7 = 34 ✓

Quick Reference Summary

KC	Topic	Core Rule / Formula	Watch Out For
KC 1	Number Patterns	Find the difference between terms; if differences grow, find 2nd-order pattern	Differences that increase are NOT constant — use second differences
KC 2	Odd & Even	Check only the last digit; Even+Even=Even, Odd+Odd=Even, Even+Odd=Odd	The product of any number with an even number is always even
KC 3	Divisibility	Div by 3/9: sum digits. Div by 2/5/10: last digit only	Divisible by 6 = divisible by both 2 AND 3
KC 4	Sequences	Arithmetic: a+(n−1)d. Geometric: a×r^(n−1). Oblong: n(n+1)	Geometric sequences have a ratio, not a difference
KC 5	Logic Puzzles	List all candidates; eliminate one constraint at a time	Always verify your answer against EVERY clue before finalising

Practice consistently — number sense is built through repetition and pattern recognition.